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Do Perioperative Probiotics/Synbiotics Reduce Postoperative Infection Rates Following Elective Colorectal Surgery? A Systematic Review and Meta-Analysis

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ARTICLE INFO

Article history:
Received 18 February 2025
Received in revised form
20 May 2025
Accepted 30 May 2025
Available online 27 June 2025

Keywords:
Colorectal surgery
Gut microbiome
Postoperative infections
Probiotics
Synbiotic

ABSTRACT

Introduction: Postoperative infections remain common in elective colorectal surgery (CRS). Perioperative probiotics/synbiotics have been investigated as a strategy to optimize the intestinal microbiota and reduce postoperative infections. The aim of this study was to conduct an updated systematic review and meta-analysis on the efficacy of perioperative probiotics/synbiotics on postoperative infection rates following elective CRS.

Methods: Six databases were searched on February 7, 2023, in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses 2020 guidelines (International Prospective Register of Systematic Reviews: CRD420233392754). Inclusion criteria: randomized controlled trials comparing perioperative probiotics or synbiotics in colorectal resection for malignant or benign disease in patients > 18 y, reporting data on infectious complications within 30 d of surgery. Exclusion criteria: patients undergoing acute colorectal resection, nonrandomized controlled trials, prebiotics alone used as the intervention or control. The primary outcome measure was total postoperative infection rates within 30 d of elective CRS.

Results: Twenty-eight randomized controlled trials involving 2686 participants demonstrated an association between probiotics/synbiotics and reduced rates of total postoperative infections within 30 d of elective CRS (relative risk 0.55, 95% confidence interval 0.41-0.74, $P < 0.001$).

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https://doi.org/10.1016/j.jss.2025.05.026

Conclusions: Perioperative probiotics/synbiotics are associated with almost halved total postoperative infection rates within 30 d of elective CRS. Further investigation into optimal regimens and the confounding effect of modern colorectal surgical practice is essential prior to implementation into clinical practice.

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Introduction

Postoperative infections remain common in elective colorectal surgery (CRS) despite advances in perioperative care and surgical technique.Five percent to 26% of patients develop surgical site infections (SSIs) alone following CRS.[1] Postoperative infections contribute to significant postoperative morbidity and mortality, as well as increased health care costs.[2]

The intestinal microbiota includes all bacteria, archaea, fungi, protozoa, and viruses living in the intestine.[3] CRS patients are exposed to a range of perioperative stressors: neoadjuvant chemoradiotherapy, '*nil by mouth*' status, mechanical bowel preparation (MBP), oral and intravenous antibiotics (OABs, IVABs), intestinal handling and surgical stress.[2,4] Perioperative stressors alter the composition of the intestinal microbiota, which has been implicated in postoperative infections.[5]

Probiotics are live microorganisms that provide a health benefit to the host.[6] Prebiotics are nondigestible foods that encourage the growth of specific bacteria.[7] Synbiotics are a combination of probiotics and prebiotics.[8] Advantages of probiotics include being widely available, inexpensive, and relatively safe with easy administration.[9] In the era of antimicrobial resistance and increasing costs, probiotics offer an attractive prospect for reducing postoperative infections following CRS.

Previously published systematic reviews on probiotics/synbiotics and postoperative infections following CRS have found that probiotics/synbiotics are associated with a reduction in postoperative infections.[10-16] Many of these systematic reviews have focused on patients with colorectal cancer (CRC), rather than encompassing patients undergoing CRS for colorectal disease, inclusive of cancer and noncancer cases. A theme in previous systematic reviews is the challenge in determining optimal treatment regimens.[15] Another limitation of previous systematic reviews is the omission of non-English studies. This consideration is important, given the substantial number of trials conducted in this field within China.

Another factor is the evolving context of modern CRS, marked by the implementation of Enhanced Recovery After Surgery (ERAS), increasing adoption of minimally invasive techniques, and the changing use of routine MBP and OABs.[2-4] These advancements are not only thought to decrease postoperative infection rates, but are also able to influence the intestinal microbiota.[2-4,9] These confounders must be considered in this field. Our aim was to perform an updated systematic review and meta-analysis, without language restriction, on the efficacy of perioperative probiotics/synbiotics on postoperative infection rates within 30 d of elective CRS.

Methods

Protocol and registration

This systematic review was reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidelines[17] and was prospectively registered with the International Prospective Register of Systematic reviews database (CRD420233392754). Informed consent was not obtained, as the study did not involve collection of new data from human participants.

Search strategy

A literature search was undertaken on February 7, 2023, on six databases: Medline, Embase, Cumulative Index to Nursing and Allied Health Literature Ultimate, Cochrane Library, Scopus, and Emcare. Search terms were developed with an information specialist librarian, using the following themes: (1) CRS, (2) Probiotics OR synbiotics, and (3). Postoperative infection. The full Medline search strategy is detailed in Supplementary Data 1. The search strategy was developed for Medline and adapted for other databases (Supplementary Data 1). No time-period restriction was used. Non-English full text papers were translated into English, with the assistance of a China-based translator identified through the freelance website Upwork. Abstract-only and unpublished studies were excluded. Reference lists of included studies were manually searched. Abdominal surgery randomized controlled trials (RCTs) were initially included, provided some of the surgeries were CRS. If CRS-specific outcomes were not published, authors were emailed with a request for raw data. Search records were loaded into Endnote, and duplicates were removed. Remaining records were uploaded for abstract and full-text screening to Covidence (Veritas Health Innovation, Melbourne, Australia).

Inclusion and exclusion criteria

Patients over the age of 18 y who underwent elective CRS for malignant or benign disease were included. The intervention had to involve probiotics or synbiotics administered within 30 d before or after surgery. Only RCTs that reported on at least one of the primary or secondary outcomes were included. RCTs involving major abdominal surgeries (colorectal and noncolorectal) were screened, but only those with extractable colorectal-specific outcome data were included in the final analysis. Non-RCTs, studies without available full text, and unpublished RCTs were excluded. Studies were excluded if

prebiotics alone were used as the control or intervention. Patients undergoing acute CRS were also excluded.

Data selection and management

Two independent reviewers (CP and AN) performed the abstract and full-text screening, based on the predetermined eligibility criteria. Disagreements were resolved by discussion with a senior author (AH). CP and AN developed a data extraction form on Covidence and independently extracted data from full-text publications (Supplementary Data 2). Discrepancies were discussed with AH. CP requested relevant data from authors when required. For RCTs that administered perioperative probiotics/synbiotics for > 30 d after surgery, the duration was converted to 30 d. For RCTs that provided a range in duration for administration of the intervention, data was collected using the smallest provided duration. For follow-up periods > 30 d following surgery, only data points occurring within 30 d of surgery were collected. Given the lack of consensus for standard care, this was considered to be either no preparation, MBP, or MBP + OABs.

Primary outcome

The primary outcome was total postoperative infection rates within 30 d following CRS. The decision to use a composite outcome was based on the proposed mechanisms for probiotics/synbiotics, which include increased competition for intestinal pathogens and immunomodulation. However, not all included RCTs calculated total postoperative infection rates, with some RCTs only providing data on SSI rates and omitting their data on non-SSIs such as urinary tract infection (UTI) or pneumoniae. For RCTs that did not report their total postoperative rates, we were unable to combine them into our primary outcome meta-analysis, but they were still able to be included in our some of secondary outcome meta-analyses.

Secondary outcomes

Secondary outcomes included individual infection rates within 30 d following CRS: SSIs, intra-abdominal abscesses (IAAs), anastomotic leak (AL), pneumonia, UTIs, line infections, and sepsis. We intended to use the Centers for Disease Control and Prevention definition of SSI,[18] but noted significant variation in the literature. We decided to define SSI to include superficial incisional SSI, deep incisional SSI, organ/space SSI, and '*wound infection*'. In addition, we had an additional secondary outcome for IAA, which was often reported separately but should have already been included as an SSI. We also had an additional secondary outcome for AL, despite this being an organ/space SSI. Non-SSIs included pneumonia, UTIs, line infections, and sepsis. These were defined by trial authors due to limited access to necessary clinical information. Line infections comprised both peripheral and central intravenous lines. Septicamia was categorized as sepsis, but bacteramia was not. Other secondary outcomes included length of stay in hospital (days), duration of postoperative antibiotics (days), time to first bowel motion (days), ileus, and mortality within 30 d following CRS.

Subgroup analyses

Preplanned subgroup analyses were performed to compare probiotics versus synbiotics and single-strain versus multi-strain formulations.

Bias assessment

CP and AN appraised studies critically using the Cochrane Collaboration Risk of Bias (ROB) tool on Covidence. The ROB summary graph and funnel plot were formulated using Review Manager 5.4.1 (Copenhagen, Denmark, Cochrane Collaboration).

Data analysis

Continuous data were visually assessed for normality using histograms. Data that were not normally distributed as presented as medians with interquartile ranges (IQR). Results were synthesized using a meta-analysis. Relative risk (RR) and 95% confidence intervals (CIs) were reported for dichotomous outcomes, and mean difference (MD) and 95% CIs for continuous outcomes. Results were displayed using a forest plot. Statistical heterogeneity was assessed by X^2 and quantified by I^2, with $I^2 > 50\%$ indicating substantial statistical heterogeneity. The Mantel–Haenszel method was used, with the fixed-effect model for outcomes with a low I^2, or the random-effects model for outcomes with a high I^2. Presence of publication bias was assessed using funnel plots. Statistical significance was set at 5%, and all tests were two-sided. *P* values below 0.001 were presented as < 0.001. Statistical analyses were performed with Review Manager 5.4.1. A prediction interval for the primary outcome was calculated with the 'metafor' package using R version 4.4.1.[19]

Results

Systematic review

After removing duplicates, 582 abstracts were screened. Forty-seven full-text reviews were performed, and 28 studies were included in the final analysis.[20-47] Two authors provided raw data that allowed inclusion of the studies in the final analysis.[24,30] One abdominal surgery RCT had colorectal-specific data that was able to be included.[24] The Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 flow diagram can be viewed in Figure 1.

The total number of participants was 2686. Of these, 1273 received probiotics or synbiotics, and 1296 received placebo or standard care. Fourteen studies utilized a placebo in the control group, whereas 14 studies used standard care. Thirteen studies used double-blinding, two used single-blinding, and 13 did not comment on the use of blinding. Included RCTs were published from 2007 to 2022. Twelve of the 28 studies were published from China, with five of these not published in English. There was a median of 74 participants in each RCT (IQR 60-96). The median of the provided mean participant ages was 64.8 y (IQR 61.0-66.5) in the intervention group, and 64.5 y (61.4-66.3) in the control group. The median

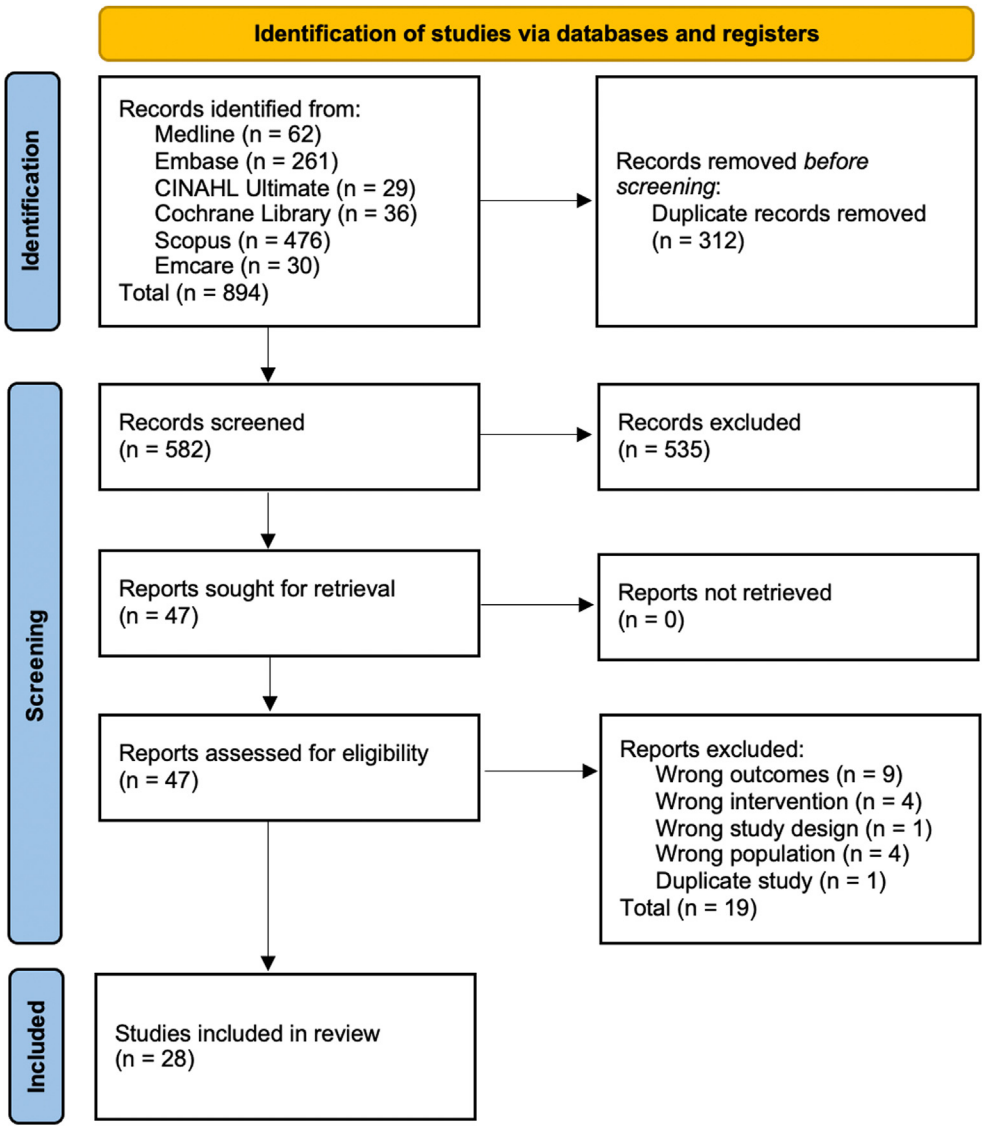

Fig. 1 – PRISMA flow diagram outlining the process of study selection to identify 28 RCTs included for final analysis. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCTs = randomized controlled trials.

percentage of females in the intervention group was 47.5%, and the control group 45.0%. Detailed study characteristics can be viewed in Table 1.

Trial regimens are available in Table 2. Twenty-two studies used probiotics, and six used synbiotics. Six studies used single-strain probiotics, and 16 used multistrain probiotics (> 2 strains). All synbiotic studies utilized multistrain probiotics. Doses ranged from 10^7 to 10^{11} colony forming units. Seven studies did not describe the doses used. *Lactobacillus* and *Bifidobacterium* were the most common genera. *Lactobacillus acidophilus* was the most common species (Fig. 2). Strain-specific probiotic information was reported in 15 of 28 studies. Types of prebiotics used included fructo-oligosaccharide, galacto-oligosaccharide, oligofructose beta-glycan, inulin, pectin, and resistant starch. Prebiotic doses ranged from 2.5g to 15g. Eleven studies administered the intervention preoperatively, three administered the intervention postoperatively, and 14 administered the intervention preoperatively and postoperatively. Duration ranged from 3 d to 1 y. Three studies did not clearly specify duration. Eleven studies used once-daily dosing, eight used twice-daily dosing, and nine used three-times-daily dosing.

Twenty-one studies used MBP in the intervention group, three studies did not, and it was unclear in four studies. Twenty-four studies used MBP in the control group, and it was unclear in four studies. Six studies described using OABs in the intervention group, six studies did not use OABs, and it was unclear in 16 studies. Eight studies described using OABs in the control group, four studies did not, and it could not be determined in 16 studies. Of the OABs used, reported types included neomycin, gentamicin + tinidazole, kanamycin + metronidazole, gentamicin + metronidazole,

Table 1 – Patient characteristics across included RCTs, and use of neoadjuvant CRT, MBP, OABs, and IVABs.

RCT	Country (no. Participants)	Surgery	Neoadjuvant therapy	MBP		OABs			IVABs		
		Pathology		Pro/syn	Control	Pro/syn	Control	Timing	Pro/syn	Control	Timing
Bajramagic 2019	Bosnia & Herzegovina ($n = 78$)	Open CRS	Excl. CRT	-	-	-	-		-	-	
		Cancer									
Chen 2014	China ($n = 60$)	Open CRS	Excl. CRT	Y	Y	-	-		-	-	
		Cancer									
Consoli 2016	Brazil ($n = 33$)	CRS	-	Y	Y	-	-		Y	Y	0.5 h preop → 1/7
		All disease							(Gen + Met)		
Flesch 2017	Brazil ($n = 91$)	CRS	Excl. CRT	Y	Y	-	-		Y	Y	1h preop
		Cancer							(Gen + Met)		
Franko 2019	USA ($n = 95$)	CRS	Excl. CT (3/52 preop)	-	-	-	-		-	-	
		All disease									
Horvat 2010	Slovenia ($n = 68$)	CRS	-	N	Y	-	-		-	-	
		Cancer									
Kakaei 2019	Iran ($n = 100$)	CRS	Excl. CT	Y	Y	Y	Y	1/7 preop TDS	Y	Y	0.5 h preop → 2/7
		All disease				(Ery + Met)			(Cefa)		
Komatsu 2016	Japan ($n = 362$)	Lap CRS	-	Y	Y	N	N		Y	Y	1h preop → 1/7
		All disease							(Ceph)		
Kotoc 2014	Czech Republic ($n = 45$)	CRS	-	Y	Y	-	-		Y	Y	3 periop doses
		Cancer							(Aug + Met)		
Kotzampassi 2015	Greece ($n = 164$)	Open CRS	Excl. CRT	Y	Y	-	-		-	-	
		Cancer									
Krebs 2013	Slovenia ($n = 54$)	CRS	-	N	Y	-	-		-	-	
		Cancer									
Li 2016	China ($n = 80$)	CRS	Excl. CRT	Y	Y	N	Y	3/7 preop TDS	Y	Y	1 dose preop
		Cancer					(Kan + Met)		(Cefop + Sul)		
Liu 2011	China ($n = 100$)	Open CRS	Excl. CRT	Y	Y	-	-		Y	Y	1h preop → 2/7
		Cancer							(Ceftr + Met)		
Liu 2013	China ($n = 150$)	CRS	Excl. CRT	Y	Y	-	-		Y	Y	1h preop → 2/7
		Cancer							(Ceftr + Met)		

(continued)

Table 1 – (*continued*)

RCT	Country (no. Participants)	Surgery	Neoadjuvant therapy	MBP		OABs			IVABs		
		Pathology		Pro/syn	Control	Pro/syn	Control	Timing	Pro/syn	Control	Timing
Mangell 2012	Sweden ($n = 64$)	Open CS	-	Y	Y	-	-		Y	Y	1 dose preop
		All disease							(Cefu + Met)		
Mizuta 2016	Japan ($n = 60$)	CRS	~25% CRT	Y	Y	-	-		Y	Y	1 dose preop → 1 dose postop day 1
		Cancer							(Cefm)		
Polakowski 2019	Brazil ($n = 73$)	CRS	Excl. CRT	Y	Y	N	N		Y	Y	Induction → 1/7
		Cancer							(Ceftr + Met)		
Reddy 2007	UK ($n = 88$)	CRS	Excl. CRT	Y (G 3)	Y	Y	Y (G 2)	1/7 preop TDS	-	-	
		All disease				(Neo)					
Sadahiro 2014	Japan ($n = 294$)	CS	-	Y	Y	N	N		Y	Y	1h preop
		Cancer							(Flo)		
Shen 2022	China ($n = 160$)	Lap CRS	Excl. CRT	N	Y	N	Y	3/7 preop TDS	-	-	
		Cancer					(Gen + Tin)				
Tan 2016	Malaysia ($n = 40$)	CRS	-	-	-	-	-		-	-	
		Cancer									
Wang 2011	China ($n = 80$)	CRS	Excl. CRT	Y	Y	Y	Y	3/7 preop	Y	Y	1 dose preop → up to 5/7
		Cancer				(Gen + Met)			(Cefu + Met)		
Xia 2008	China ($n = 60$)	CRS	Excl. CRT	Y	Y	Y	Y	3/7 preop TDS	Y	Y	Intraop → 3/7
		Cancer				(Gen + Met)			(Cefu)		
Xia2010	China ($n = 60$)	CRS	Excl. CRT	Y	Y	-	-		Y	Y	Intraop → 3/7
		Cancer							(Cefu)		
Xu 2019	China ($n = 60$)	CRS	-	-	-	-	-		-	-	
		Cancer									
Yang 2014	China ($n = 75$)	CRS	-	Y	Y	Y	Y	3/7 preop TDS	Y	Y	Preop → 3/7
		Cancer				(Kan + Met)			(Cefop + Sul)		
Yang 2016	China ($n = 60$)	Open CRS	Excl. CRT	Y	Y	-	-		Y	Y	0.5 h preop
		Cancer							(Cefox)		
Zhang 2012	China ($n = 60$)	Open CRS	Excl. CRT	Y	Y	Y	Y	2/7 preop TDS	Y	Y	Intraop → up to 5/7
		Cancer				(Gen + Met)			(Gen + Met)		

CRS = colorectal surgery; lap = laparoscopic; CS = colonic surgery; CRT = chemoradiotherapy; CT = chemotherapy; Y = yes (administered); N = Not administered; - = Not stated; G = group; Gen = gentamicin; Met = metronidazole; Cefa = cefazolin; Ery = erythromycin; Ceph = cephalosporin (not specified); Aug = augmentin; Cefop = cefoperazone; Sul = sulbactam; Cefm = cefmetazole; Ceftr = ceftriaxone; Neo = neomycin; Flo = flomoxef; Tin = tinidazole; Kan = kanamycin; Cefox = cefoxitin; preop = preoperative; periop = perioperative; intraop = intraoperative; RCTs = randomized controlled trials; MBP = mechanical bowel preparation; OABs = oral antibiotics; IVABs = intravenous antibiotics.

Table 2 – Probiotic/synbiotic regimens utilized in included RCTs.

Study	Type	Strain	Probiotic content (CFU, if available) / Prebiotic content (dose in g)	Frequency	Preop (d)	Postop (d)
Bajramagic 2019	Pro	Multi	*L. acidophilus*, *L. casei*, *L. plantarum*, *L. rhamnosus*, *B. lactis*, *B. bifidum*, *B. breve*, *S. thermophilus*	BD	-	30
Chen 2014	Pro	Multi	*C. butyricum*, *B. infantis*	TDS	5	7
Consoli 2016	Pro	Single	*S. boulardii* (0.5×10^9)	OD	7	-
Flesch 2017	Syn	Multi	*L. acidophilus*, *L. rhamnosus*, *L. paracasei*, *B. lactis* (all 10^8-10^9)	BD	5	14
			Fructooligosaccharide (6g)			
Franko 2019	Pro	Multi	*B. breve*, *B. longum*, *B. infantis*, *L. acidophilus*, *L. plantarum*, *L. paracasei*, *L. bulgaricus*, *S. thermophilus* (all 1.25×10^{10})	BD	1	13
Horvat 2010	Syn	Multi	*P. pentosaceus*, *L. mesenteroides*, *L. paracasei*, *L. plantarum* (all 10^{10})	BD	3	-
			Beta-glycan, inulin, pectin, resistant starch (all 2.5 g)			
Kakaei 2019	Pro	Multi	*L. casei*, *L. acidophilus*, *L. plantarum*, *B. breve*, *B. longum*, *S. thermophilus* (all 10^9)	OD	7	23
Komatsu 2016	Syn	Multi	*L. casei* (4×10^{10}), *B. breve* (10^{10})	OD	7	?
			Galactooligosaccharide (2.5 g)			
Kotoc 2014	Pro	Multi	*L. rhamnosus*, *B. breve*, *L. casei*, *L. acidophilus*, *B. longum* (all 6×10^9)	BD	10	-
Kotzampassi 2015	Pro	Multi	*L. acidophilus* (1.75×10^9), *L. plantarum* (0.5×10^9), *B. lactis* (1.75×10^9), *S. boulardii* (1.5×10^9)	BD	1	14
Krebs 2013	Syn	Multi	*P. pentosaceus*, *L. mesenteroides*, *L. paracasei*, *L. plantarum* (all 10^{11})	OD	3	-
			Beta-glycan, inulin, pectin, resistant starch (all 2.5 g)			
Li 2016	Pro	Multi	*Lactobacillus bifidum triptans*	TDS	5	7
Liu 2011	Pro	Multi	*L. plantarum* ($>10^{11}$), *L. acidophilus* ($>7 \times 10^{10}$), *B. longum* ($>5 \times 10^{10}$)	OD	6	10
Liu 2013	Pro	Multi	*L. plantarum* ($>10^{11}$), *L. acidophilus* ($>7 \times 10^{10}$), *B. longum* ($>5 \times 10^{10}$)	OD	6	10
Mangell 2012	Pro	Single	*L. plantarum* (10^{11})	OD	8	5
Mizuta 2016	Pro	Single	*B. longum* (5×10^{10})	OD	7-14	14
Polakowski 2019	Syn	Multi	*L. acidophilus*, *L. rhamnosus*, *L. casei*, *B. lactis* (all 10^9)	BD	7	-
			Fructooligosaccharide (6g)			
Reddy 2007	Syn	Multi	*L. acidophilus*, *L. bulgaricus*, *B. lactis*, *S. thermophilus* (all 4×10^9)	TDS	?	-
			Oligofructose (15g)			
Sadahiro 2014	Pro	Single	*B. bifidum* ($10^{10)}$	TDS	7	10
Shen 2022	Pro	Multi	*Lactobacillus bifidum triptans*	TDS	3	-
Tan 2016	Pro	Multi	*L. acidophilus*, *L. casei*, *L. lactis*, *B. bifidum*, *B. longum*, *B. infantis* (all 30×10^{10})	BD	7	-
Wang 2011	Pro	Multi	*B. infantis*, *L. acidophilus*, *E. faecalis*, *B. cereus*	TDS	5	-
Xia 2008	Pro	Single	*L. acidophilus* ($6\text{-}10 \times 10^{8)}$	OD	-	7
Xia 2010	Pro	Single	*L. acidophilus* ($6\text{-}10 \times 10^8$)	OD	5	-
Xu 2019	Pro	Multi	*Lactobacillus bifidum triptans*	OD	-	7
Yang 2014	Pro	Multi	*B. longum*, *L. bulgaris*, *S. thermophilus*	TDS	5	?
Yang 2016	Pro	Multi	*B. longum*, *L. acidophilus*, *E. faecalis* ($>10^7$)	TDS	5	7
Zhang 2012	Pro	Multi	*B. longum*, *L. acidophilus*, *E. faecalis* (10^8)	TDS	3	-

Pro = probiotics; Syn = synbiotics; CFU = colony forming units; OD = once-daily; BD = twice-daily; TDS = three-times-daily; preop = preoperative; postop = postoperative; - = not administered; ? = not stated; RCTs = randomized controlled trials.

and erythromycin + metronidazole. Eighteen studies described using IVABs, and it was unclear in ten studies. The most common IVABs regimen was a cephalos-?tjl] porin + metronidazole. Full results are demonstrated in Table 1. Detailed multi-resistant organism (MRO) status and culture isolates were not frequently reported. One study reported on methicillin-resistant *Staphylococcus aureus* infection rates after CRS. One study provided bacterial culture isolates following incisional SSIs, and a second study provided bacterial culture isolates following postoperative infections.

Twenty studies did not specify the surgical approach. Two studies included only minimally invasive CRS, and six studies

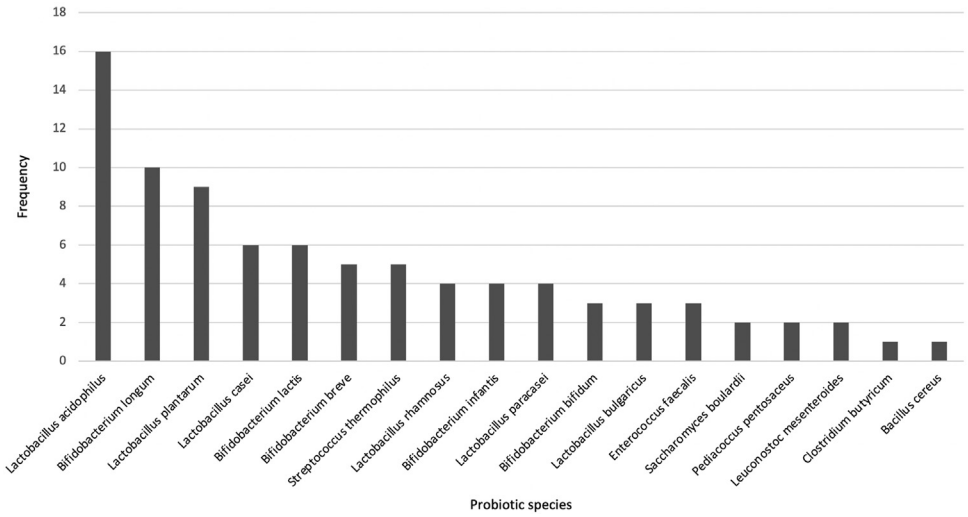

Fig. 2 – Probiotics species utilized in included RCTs. *Lactobacillus acidophilus* was the most frequently utilized, appearing in 16 formulations. RCTs = randomized controlled trials.

included only open CRS. Twenty-one studies included CRC patients, and five studies included patients with colorectal disease. One study included colon cancer only, and another included colon disease only. One study reported that a proportion of participant had neoadjuvant chemoradiotherapy (CRT); the majority of studies excluded patients who had neoadjuvant CRT. ERAS was mentioned in only three RCTs and was first mentioned in 2016. Full results can be visualized in Table 1.

Meta-analysis

Data on the primary outcome of total postoperative infection rates within 30 d of CRS was reported in 14 of 28 RCTs, with 1270 participants (Fig. 3). Pooled data demonstrated a significant reduction in total postoperative infection rates in the intervention group compared with the control group (RR 0.55, 95% CI 0.41-0.74, $P < 0.001$, $I^2 = 51\%$). The 95% prediction interval was 0.26-1.15, indicating a wide range of plausible values for the true effect size in a new, similar study, accounting for sampling variability and heterogeneity.

Data on the secondary outcomes of individual infection rates within 30 d of CRS were reported more commonly than the primary outcome across the 28 included RCTs. A significant reduction in pneumonia was demonstrated in the intervention group compared to control (RR 0.39, 95% CI 0.26-0.59, $P < 0.001$, $I^2 = 0\%$) among 17 studies with 1353 participants (Fig. 4). UTIs were significantly decreased in the intervention group compared with control (RR 0.49, 95% CI 0.29-0.82, $P = 0.007$, $I^2 = 0\%$) among nine studies with 716 participants (Fig. 5). A significant reduction in reported SSIs were seen in the intervention group compared with control (RR 0.60, 95% CI 0.48-0.75, $P < 0.001$, $I^2 = 0\%$) among 24 studies with 2191 participants (Fig. 6). There was a significant reduction in IAAs

Study or Subgroup	Pro/Synbiotics Events	Pro/Synbiotics Total	Placebo/Standard Care Events	Placebo/Standard Care Total	Weight	Risk Ratio M-H, Random, 95% CI
Bajramagic 2019	18	39	26	39	13.6%	0.69 [0.46, 1.04]
Chen 2014	23	30	27	30	16.4%	0.85 [0.68, 1.07]
Consoli 2016	2	15	7	18	3.5%	0.34 [0.08, 1.41]
Franko 2019	5	41	10	54	5.9%	0.66 [0.24, 1.78]
Kakaei 2019	4	50	7	50	4.7%	0.57 [0.18, 1.83]
Kotzampassi 2015	10	84	23	80	9.3%	0.41 [0.21, 0.81]
Li 2016	2	40	9	40	3.3%	0.22 [0.05, 0.96]
Liu 2011	7	50	23	50	8.3%	0.30 [0.14, 0.64]
Mangell 2012	4	32	6	32	4.7%	0.67 [0.21, 2.14]
Polakowski 2019	1	36	7	37	1.9%	0.15 [0.02, 1.13]
Reddy 2007	6	42	9	46	6.3%	0.73 [0.28, 1.88]
Sadahiro 2014	24	100	24	95	12.1%	0.95 [0.58, 1.55]
Wang 2011	4	40	12	40	5.5%	0.33 [0.12, 0.95]
Zhang 2012	3	30	10	30	4.6%	0.30 [0.09, 0.98]
Total (95% CI)		**629**		**641**	**100.0%**	**0.55 [0.41, 0.74]**
Total events	113		200			

Heterogeneity: Tau² = 0.12; Chi² = 26.54, df = 13 (P = 0.01); I² = 51%
Test for overall effect: Z = 3.95 (P < 0.0001)

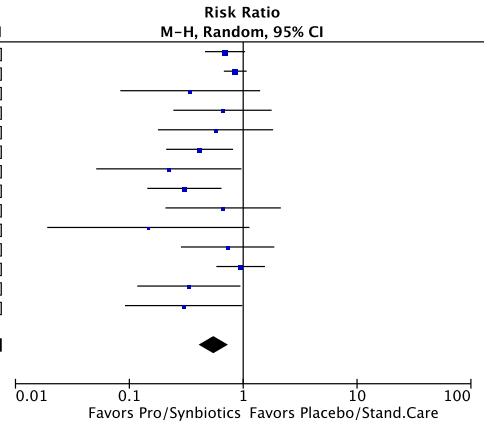

Fig. 3 – Forest plot including 14 RCTs demonstrating a significant reduction in total postoperative infections observed when probiotics/synbiotics were used compared to standard care/placebo. RCTs = randomized controlled trials.

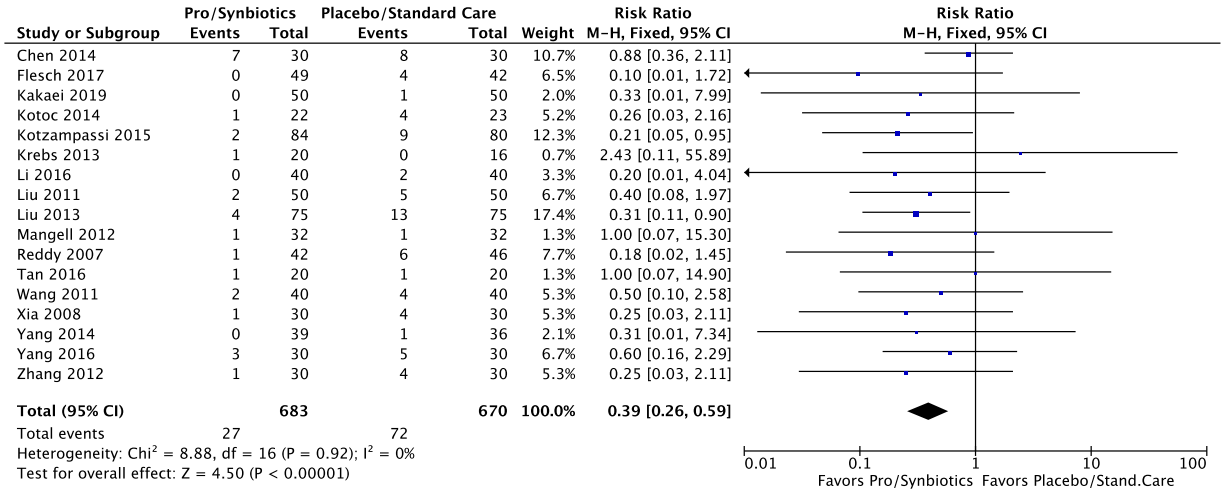

Fig. 4 – Forest plot including 17 RCTs showing a significant reduction in pneumonia observed with the use of probiotics/synbiotics. RCTs = randomized controlled trials.

among the intervention group compared to control (RR 0.52, 95% CI 0.30-0.89, $P = 0.02$, $I^2 = 0\%$) among 12 studies with 924 participants (Supplementary Data 3). No significant reduction in AL was observed in the intervention group compared with control (RR 0.74, 95% CI 0.50-1.11, $P = 0.15$, $I^2 = 1\%$) among 15 studies with 1667 participants (Supplementary Data 4). Pooled data demonstrated a significant reduction in line infections in the intervention group compared to control (RR 0.32, 95% CI 0.15-0.69, $P = 0.003$, $I^2 = 0\%$) among four studies with 390 participants (Supplementary Data 5). There were insufficient data to describe the proportion of participants with central lines *in-situ* postoperatively. There was no significant reduction in sepsis observed in the intervention group compared with control (RR 0.36, 95% CI 0.13-1.00, $P = 0.05$, $I^2 = 41\%$) among five studies with 529 participants (Supplementary Data 6).

Data on noninfectious secondary outcomes were reported with varying frequencies across the 28 RCTs. There was a statistically significant reduction in mortality observed in the intervention group compared to control (RR 0.27, 95% CI 0.08-0.97, $P = 0.04$, $I^2 = 0\%$) among 14 studies with 1249 participants, although not all studies commented on mortality (Supplementary Data 7). A significant reduction in ileus was demonstrated in the intervention group compared to control (RR 0.27, 95% CI 0.08-0.93, $P = 0.04$, $I^2 = 56\%$) among four studies with 544 participants (Supplementary Data 8). A significant reduction in length of stay in hospital was seen in the intervention group compared with control (MD −0.99 d, 95% CI −1.87 to −0.11, $P = 0.03$, $I^2 = 64\%$) among seven studies with 600 participants (Supplementary Data 9). There was no significant reduction in duration of postoperative antibiotics in the intervention group compared to control (MD −0.84 d 95% CI −2.18 to 0.50, $P = 0.22$, $I^2 = 92\%$) among five studies with 443 participants (Supplementary Data 10). No significant reduction in time to first bowel motion was seen in the intervention group compared with control (MD −0.24 d, 95% CI −1.20 to 0.73, $P = 0.63$, $I^2 = 96\%$) among six studies with 513 participants (Supplementary Data 11).

Subgroup analyses were performed, firstly comparing probiotics to synbiotics, and secondly, single-strain to

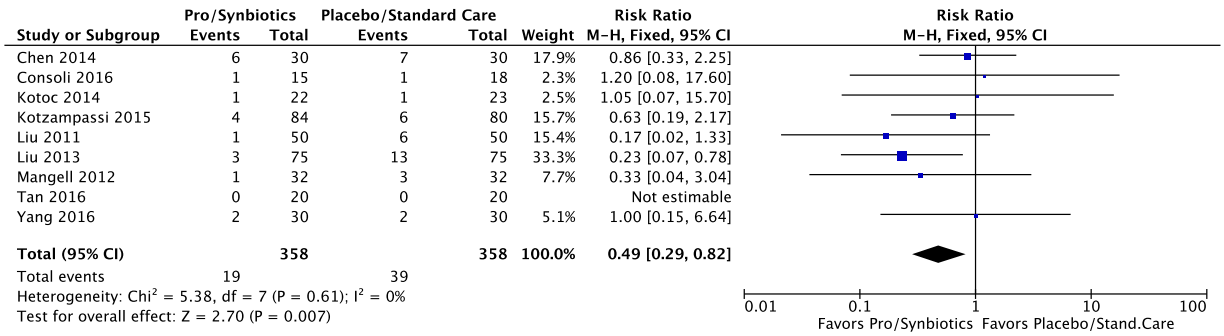

Fig. 5 – Forest plot including 9 RCTs exhibiting the significant reduction in UTIs seen with the use of probiotics/synbiotics. RCTs = randomized controlled trials; UTIs = urinary tract infections.

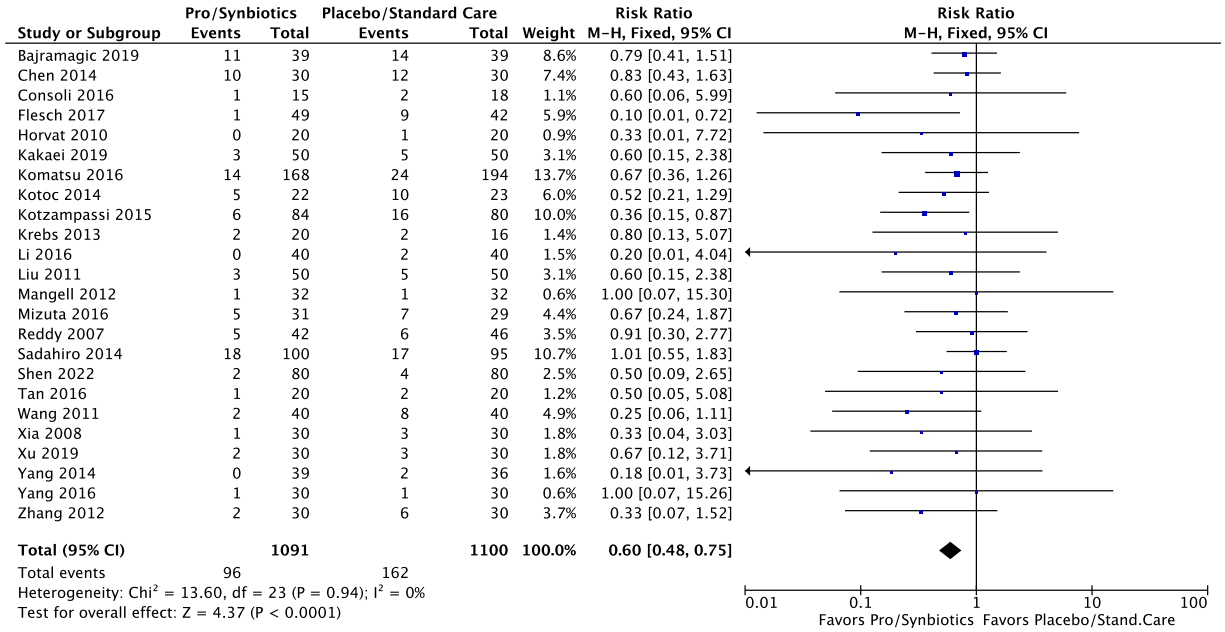

Fig. 6 – Forest plot including 24 RCTs demonstrating the significant reduction in surgical site infections observed with the perioperative administration of probiotics/synbiotics. RCTs = randomized controlled trials.

multistrain regimens. Probiotics demonstrated an RR of 0.56 (95% CI 0.41-0.76), while synbiotics showed an RR of 0.42 (95% CI 0.09-1.97). Test for subgroup differences did not achieve statistical significance ($P = 0.72$). Single strain exhibited an RR of 0.82 (95% CI 0.53-1.27), while multistrain demonstrated an RR of 0.50 (95% CI 0.34-0.72). Subgroup differences did not achieve statistical significance ($P = 0.08$).

Quality analysis

TheROB analysis results are summarized in Figure 7. There was a wide range of quality among the included RCTs, with lack of methodological details raising some concerns. A funnel plot is shown in Figure 8 and did not demonstrate significant concern for publication bias.

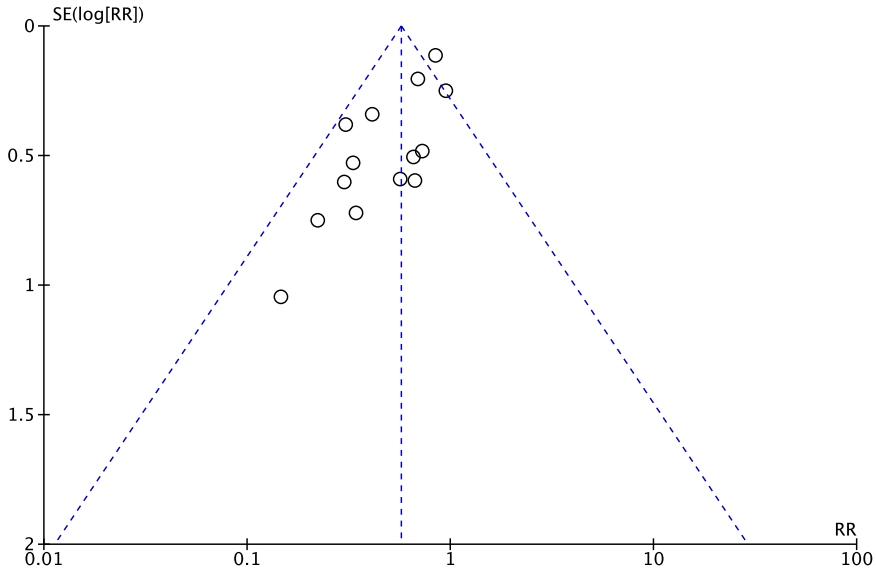

Fig. 7 – ROB summary of included RCTs, providing a visual overview of the methodological quality and potential biases within studies. ROB = risk of bias; RCTs = randomized controlled trials.

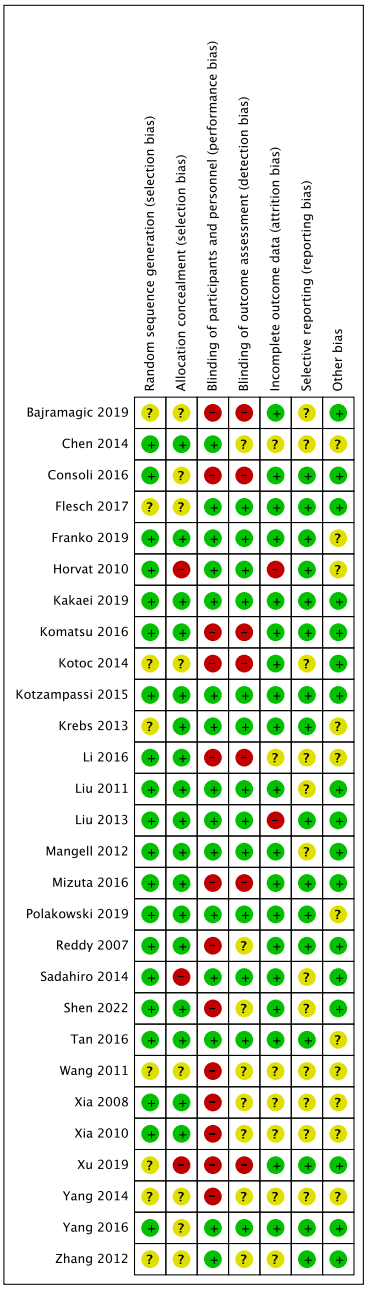

Fig. 8 – Funnel plot illustrating the relationship between the effect sizes and standard errors of the 14 RCTs included in assessing total postoperative infection rates. The relative symmetry does not cause significant concern for risk of publication bias. RCTs = randomized controlled trials.

Discussion

CRS patients are at high risk of postoperative infections, and substantial efforts have been directed toward mitigating these risks.[48] This systematic review focused on the association between perioperative probiotics/synbiotics and postoperative infection rates within 30 d of CRS. We observed significant reductions in total postoperative infections, pneumonia, UTIs, SSIs, IAAs, and line infections.

Included RCTs were conducted across continents, with many of these originating from China. A majority of participants had CRC, rather than noncancerous disease. Sixteen of 28 RCTs failed to demonstrate statistically significant reductions in infectious complications.[21,22,24-27,30,34,35,37,38,40,42-44,46] Many of these RCTs had small sample sizes, which may have been underpowered. We believe that the findings from this systematic review are generalizable to adult patients undergoing CRS for cancer and noncancerous disease. Other systematic reviews have concluded that synbiotics are likely more effective than probiotics, and multistrain probiotics are likely more effective than single-strain, but contrastingly, our subgroup analyses failed to demonstrate this.[11,49] Small sample sizes were seen across multiple subgroups, and interpretations were again limited by lack of statistical power.

The most recent systematic review in this field was published in 2024 and only included 12 studies. The authors found similar reductions in total postoperative infection rates, SSIs, pneumonia, and UTIs.[16] They did not review sepsis, line infections, or ileus rates. Our work found that probiotics/synbiotics significantly reduced ileus rates. A network meta-analysis published in 2023 on probiotics/synbiotics and ileus rates following all types of abdominal surgery also found that probiotics were associated with decreased rates of ileus.[50] Reduction of this common noninfectious complication is of relevance to CRS patients and warrants further investigation. This systematic review did not show a decrease in AL, which is an important finding to note when considering recent work that has shown that pathogenic bacteria such as collagenase-producing *E. faecalis* are able to contribute to development of AL.[4]

Another systematic review was published 3 y prior to our work and included 21 RCTs—albeit restricted included articles to those published in English and French.[15] The restriction of non-English articles is problematic when many of the relevant articles originate from China. Our systematic review sought to include as many relevant studies as possible despite language barriers, providing a nonbiased and global perspective. Their systematic review found that probiotics/synbiotics reduce total postoperative infections following CRS and noted a specific reduction in non-SSIs such as pneumonia and UTIs.[15] Our systematic review revealed similar findings, with the addition of finding a statistically significant reduction in mortality rates. Further work is required before establishing this as a true finding, as opposed to a type 1 error. We also attempted to highlight the importance on describing MRO status and specific choice of antibiotic, which is a unique contribution to this field. Although this was poorly reported, this concept is of increasing importance in the antimicrobial resistance era, with the ubiquitous use of prophylactic antibiotics.

Multiple proposed strain-specific mechanisms of actions exist for probiotics.[51] Viable probiotic bacteria in the lower gastrointestinal tract enhance microbial diversity, competing against pathogens.[52] Probiotics play a role in producing short-chain fatty acids such as butyrate, which provide energy for colonic epithelial cells.[53] Probiotics are thought to improve the intestinal barrier by promoting the production of tight-

junction proteins such as zonulin and encouraging mucous production, which decreases bacterial translocation from the lumen to mesenteric lymph nodes and the systemic circulation.[51,54] Lastly, probiotics are thought to have immunomodulatory effects.[55] These proposed mechanisms may assist in perioperative optimization of the intestinal microbiota, reducing the risk of postoperative infections.

A multitude of factors have been found to affect the intestinal microbiota, such as diet, geography, genetics, disease status, and medications. Diet is considered to be a major factor, and is also rapidly modified.[56] Jain and colleagues compared fecal diversity in healthy Indian and Chinese individuals and found a high abundance of the genus *Bifidobacterium*, thought to be associated with a carbohydrate-based Asian diet.[57] Contrastingly, a Western diet (characterized by high saturated fats and low fiber), is associated with decreased levels of *Bifidobacterium*.[57] Further work is required to elucidate the significance of varying levels of specific bacteria across dietary differences and how this could be targeted to improve outcomes in CRS patients.

Major potential confounders include use of MBP, OABs, IVABs, and neoadjuvant CRT. These have not been given sufficient attention in previous systematic reviews. The majority of studies excluded patients who had undergone neoadjuvant CRT, which eliminated this as a major confounder. However, MBP, OABs, and IVABs remained as major confounders.[58] Our literature search identified a number of studies that lacked sufficient variation in both use and reporting of use with respect to MBPs, OABs, and IVABs. This was outlined in detail in Table 1, which provides a visual interpretation of the lack of information available.

We acknowledge that our study has multiple limitations. Most notably, the lack of available information on MBP and OABs weakens the strength of the conclusions able to be drawn from the meta-analysis. It is vital that future researchers provide comprehensive details on the use of MBP and OABs to facilitate meaningful progress in understand the role of the gut microbiome in reducing postoperative infection rates following elective CRS. Specifically, RCTs should clarify whether MBP or OABs have been used in the intervention and control groups, and if OABs have been used, the specific type of OABs should be clearly documented.

Future studies should aim to better characterize the following in more detail: individual variations in intestinal microbiota; comorbidities; location of disease; the presence of cancer or noncancerous disease; surgical approach; and adherence to ERAS protocols. We recommend that postoperative infections should be monitored for 30 d following surgery, inclusive of inpatient and community surveillance. Severity of postoperative complications should be quantified, using tools such as the Comprehensive Complication Index.[59] SSIs should be categorized into superficial incisional SSI, deep incisional SSI, and organ/space SSI.[18] Furthermore, few RCTs specified MRO status or bacterial isolates from cultures. To provide effective interventions against SSIs, data on specific bacterial pathogens is crucial.[60] Lastly, details regarding the following probiotic/synbiotic regimens should also be clearly documented in future trials: dose, frequency, route of administration, duration, perioperative timing, strain selection, use of single-strain or multistrain probiotics, and use of probiotics or synbiotics.

Conclusions

In conclusion, this systematic review demonstrated that perioperative probiotics/synbiotics almost halve total postoperative infection rates following CRS. We are not yet at the stage where we can fully utilize this benefit for individual patients requiring CRS. Future research should focus on heterogeneity at the patient-, intervention-, and outcome-level.

Supplementary Materials

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jss.2025.05.026.

CRediT authorship contribution statement

Claudia Paterson: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amanda Nikolic:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Tamara Glyn:** Writing – review & editing, Supervision, Methodology. **Tim Eglinton:** Writing – review & editing, Supervision, Methodology. **Parry Singh:** Writing – review & editing, Supervision, Methodology. **Andrew Hill:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Acknowledgments

We thank the two trial authors who responded to data requests, our information specialist librarian, and our translator.

Disclosure

None declared.

Funding

Claudia Paterson is funded by the Health Research Council of New Zealand on a Clinical Research Training Fellowship. The authors have no conflicts of interest to disclose.

Meeting Presentation

This work was presented orally at the 20th Annual Academic Scientific Congress in Las Vegas, Nevada on February 11th, 2025.

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